PROGRAM CODE

```
colors = ['Red', 'Green', 'Blue']

graph = {
    'A': ['B', 'C'],
    'B': ['A', 'C'],
    'C': ['A', 'B']
}

result = {}

def color_map(node):
    if node not in result:
        for color in colors:
            if all(result.get(n) != color for n in graph[node]):
                result[node] = color
                if len(result) == len(graph):
                    return True
                if color_map(next(n for n in graph if n not in result)):
                    return True
                del result[node]
        return False
    return True

color_map('A')
print(result)
```

OUTPUT

```
PS D:\AI LAB Vijay> & C:/Users/SAIL/AppData/Loca
Vijay/Exp11"
{'A': 'Red', 'B': 'Green', 'C': 'Blue'}
PS D:\AI LAB Vijay>
```